

Geometry
Unit 2
Lesson 8 Homework

Name _____
Date _____
Period _____

Complete the two-column proof.

Given: $\overline{AB} \cong \overline{CD}$

Prove: $\overline{AC} \cong \overline{BD}$



	Statement	Reason
1.	$\overline{AB} \cong \overline{CD}$	Given
2.	$\overline{BC} \cong \overline{BC}$	Reflexive Property
3.	$AB = CD$ $BC = BC$	Definition of Congruence
4.	$AB + BC = AC$ $CD + BC = BD$	Segment Addition Postulate
5.	$CD + BC = AC$	Substitution Property
6.	$AC = BD$	Substitution Property
7.	$\overline{AC} \cong \overline{BD}$	Definition of Congruence

A partially completed two-column proof is shown. Match each statement on the left with the correct reason to complete the two-column proof by using reasons provided in the bank. Some reasons may be used more than once.

Given: W is the midpoint of $\overline{VX}$, Z is the midpoint of $\overline{RP}$, and $\overline{VX} \cong \overline{RP}$

Prove: $\overline{WX} \cong \overline{ZP}$



Reasons			
Given	Combine like terms	Definition of congruence	Definition of midpoint
Division property of equality		Segment addition postulate	Substitution property of equality

	Statement	Reason
1.	W is the midpoint of $\overline{VX}$, Z is the midpoint of $\overline{RP}$, and $\overline{VX} \cong \overline{RP}$	Given
2.	$\overline{VW} \cong \overline{WX}$ $\overline{RZ} \cong \overline{ZP}$	Definition of midpoint
3.	$VX = RP$	Definition of congruence
4.	$VW + WX = VX$ $RZ + ZP = RP$	Segment addition postulate
5.	$VW = WX$ $RZ = ZP$	Definition of congruence
6.	$VW + WX = RZ + ZP$	Substitution property of equality
7.	$WX + WX = ZP + ZP$	Substitution property of equality
8.	$2WX = 2ZP$	Combine like terms
9.	$WX = ZP$	Division property of equality
10.	$\overline{WX} \cong \overline{ZP}$	Definition of congruence